

Day 5: Configuring OSPF

Task 1. Create a network topology with three routers in two (or more) OSPF areas, and assign appropriate IP addresses to all interfaces.

A topology of three routers (R1, R2, R3) was built in Cisco Packet Tracer, fully meshed over serial links so that all three sit in the OSPF backbone, Area 0. Each router also connects to its own LAN, placed in a separate area, making each router an ABR (Area Border Router) between Area 0 and its local area. This goes beyond the minimum two-area requirement and demonstrates a four-area design (Area 0 backbone plus Areas 1, 2, and 3).

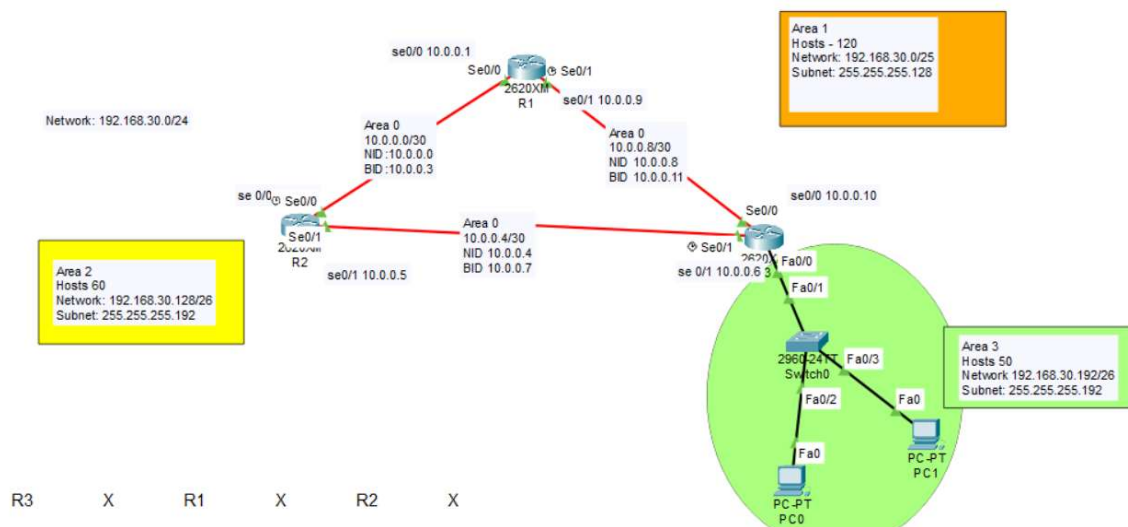


Figure 1: Topology - R1, R2, R3 in Area 0 (backbone), with Area 1, Area 2, and Area 3 LANs hanging off each router.

IP addressing plan:

Area	Network	Subnet Mask	Purpose
Area 0 (Backbone)	10.0.0.0/30, 10.0.0.4/30, 10.0.0.8/30	255.255.255.252	R1-R2, R2-R3, R1-R3 serial links
Area 1	192.168.30.0/25	255.255.255.128	R1-side LAN, up to 120 hosts
Area 2	192.168.30.128/26	255.255.255.192	R2 LAN (PC0), up to 60 hosts
Area 3	192.168.30.192/26	255.255.255.192	R3 LAN (PC0/PC1 via switch), up to 50 hosts

Task 2. Configure OSPF on each router so Area 0 acts as the backbone and the other areas connect through it; verify neighbor relationships using 'show ip ospf neighbor'.

OSPF process 1 was enabled on all three routers, with each serial (backbone) interface placed in Area 0 using its /30 network, and each LAN interface placed in its own area (Area 1, 2, or 3) using the appropriate network statement, for example:

- router ospf 1
- network 10.0.0.0 0.0.0.3 area 0
- network <LAN subnet> <wildcard> area <n>

Running 'show ip ospf neighbor' on all three routers confirmed full adjacency across every backbone link:

Router	Router ID	Neighbors (Full Adjacency)	Interfaces
R1	1.1.1.1	2.2.2.2 (R2), 3.3.3.3 (R3)	Serial0/0, Serial0/1
R2	2.2.2.2	1.1.1.1 (R1), 3.3.3.3 (R3)	Serial0/0, Serial0/1
R3	3.3.3.3	1.1.1.1 (R1), 2.2.2.2 (R2)	Serial0/0, Serial0/1

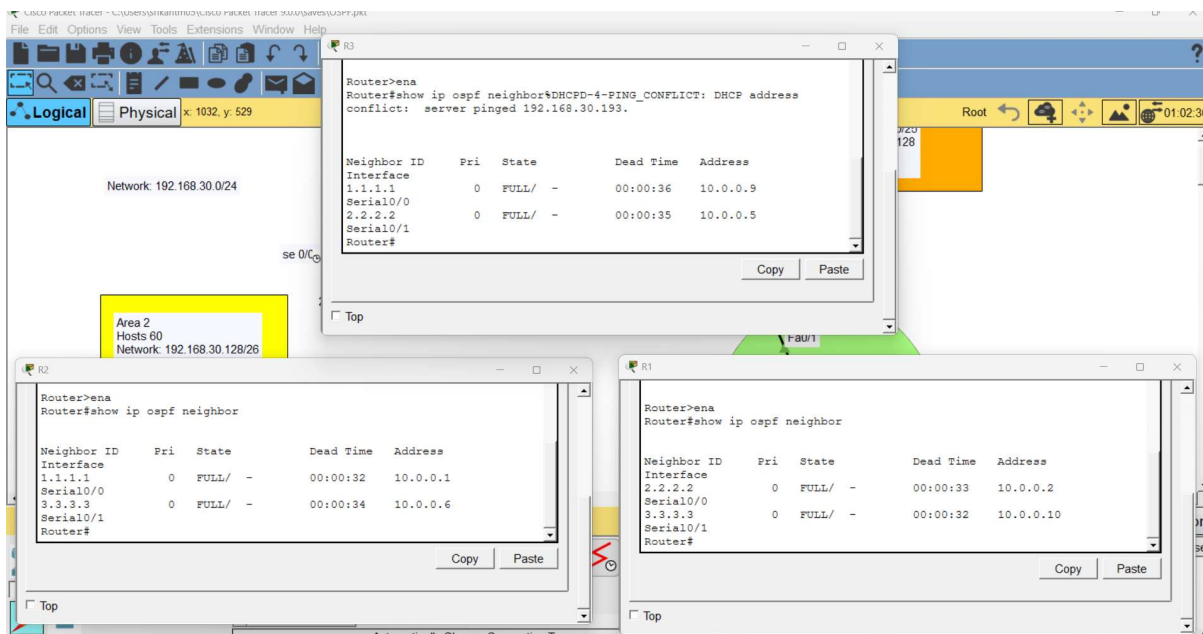
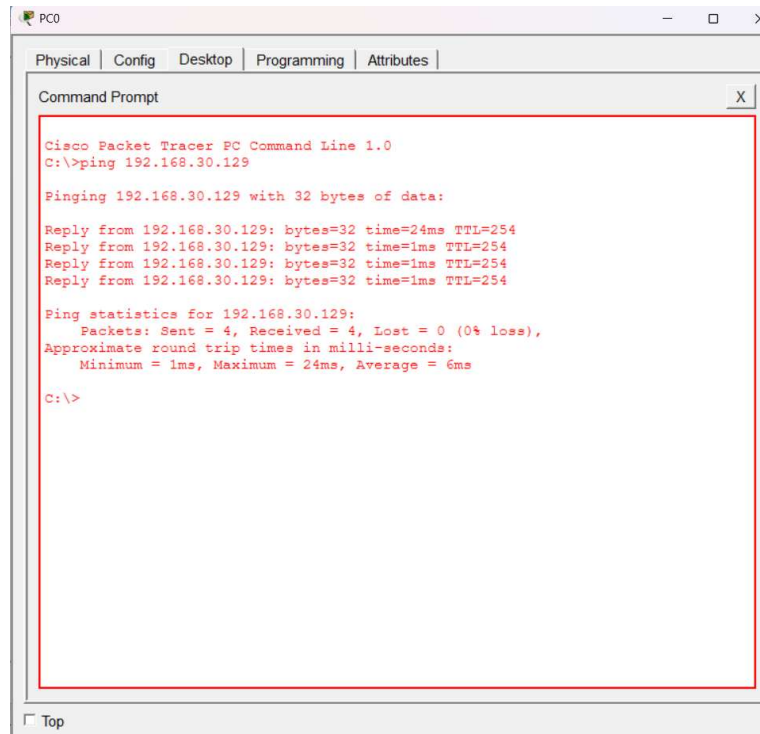


Figure 2: 'show ip ospf neighbor' output on R3, R2, and R1 - all neighbors show state FULL/-, confirming a complete Area 0 mesh.

Task 3. Add two PCs to different areas, assign them IP addresses, and test connectivity by pinging from one PC to the other.

PC0 was placed on R2's LAN (Area 2, 192.168.30.128/26) and PC1/PC2 were placed on R3's LAN (Area 3, 192.168.30.192/26), connected through a switch. Each PC was assigned a static IP address within its area's subnet, with the local router's LAN interface as the default gateway.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.129

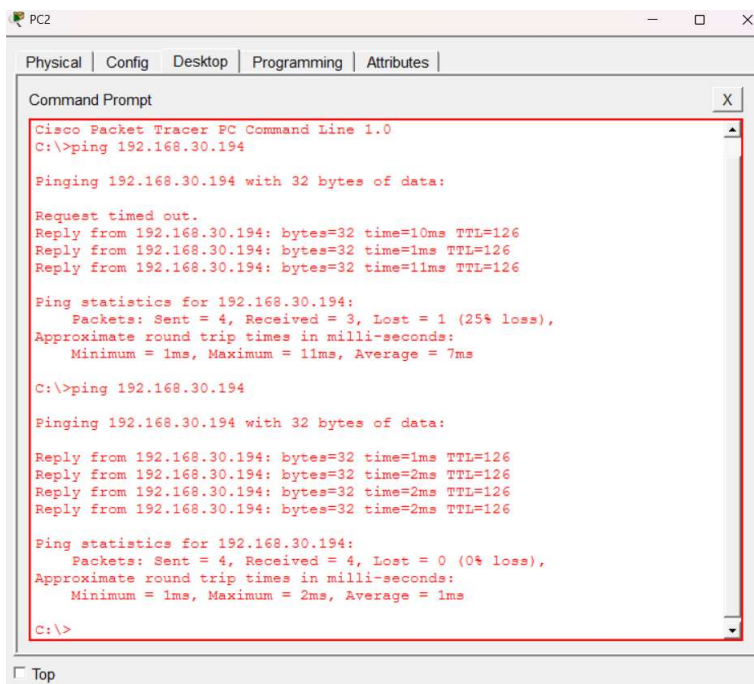
Pinging 192.168.30.129 with 32 bytes of data:

Reply from 192.168.30.129: bytes=32 time=24ms TTL=254
Reply from 192.168.30.129: bytes=32 time=1ms TTL=254
Reply from 192.168.30.129: bytes=32 time=1ms TTL=254
Reply from 192.168.30.129: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.30.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 24ms, Average = 6ms

C:\>
```

Figure 3: PC0 (Area 2) successfully pings 192.168.30.129 - 4/4 packets received, 0% loss.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.194

Pinging 192.168.30.194 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.194: bytes=32 time=10ms TTL=126
Reply from 192.168.30.194: bytes=32 time=1ms TTL=126
Reply from 192.168.30.194: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.30.194:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 7ms

C:\>ping 192.168.30.194

Pinging 192.168.30.194 with 32 bytes of data:

Reply from 192.168.30.194: bytes=32 time=1ms TTL=126
Reply from 192.168.30.194: bytes=32 time=2ms TTL=126
Reply from 192.168.30.194: bytes=32 time=2ms TTL=126
Reply from 192.168.30.194: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.30.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

Figure 4: PC2 (Area 3) successfully pings 192.168.30.194 across areas

Both tests confirm end-to-end, inter-area reachability: OSPF correctly installed 'O IA' (inter-area) routes on each router for the remote LANs, and traffic was routed across the Area 0 backbone without issue.

Task 4. Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf neighbor' to observe reconvergence after re-enabling the interface.

The direct serial link between R1 and R2 (part of the Area 0 backbone mesh) was shut down to simulate a link/router failure. 'show ip ospf neighbor' and 'show ip route' were then re-checked on all three routers:

- R1: the neighbor relationship with R2 (2.2.2.2) disappeared - only R3 (3.3.3.3) remained as a FULL neighbor.
- R2: the neighbor relationship with R1 (1.1.1.1) disappeared - only R3 (3.3.3.3) remained as a FULL neighbor.
- R3: unaffected - it retained FULL adjacencies with both R1 and R2, since it was not on the failed link.
- R1's routing table showed its route to R2's backbone subnet and LAN (192.168.30.128/26) had shifted to go via R3 (10.0.0.10, Serial0/1) instead of the direct link, with a higher OSPF cost (110/129 versus the earlier lower-cost direct path).
- R2 similarly showed its route to R1's side of the network re-routed via R3.

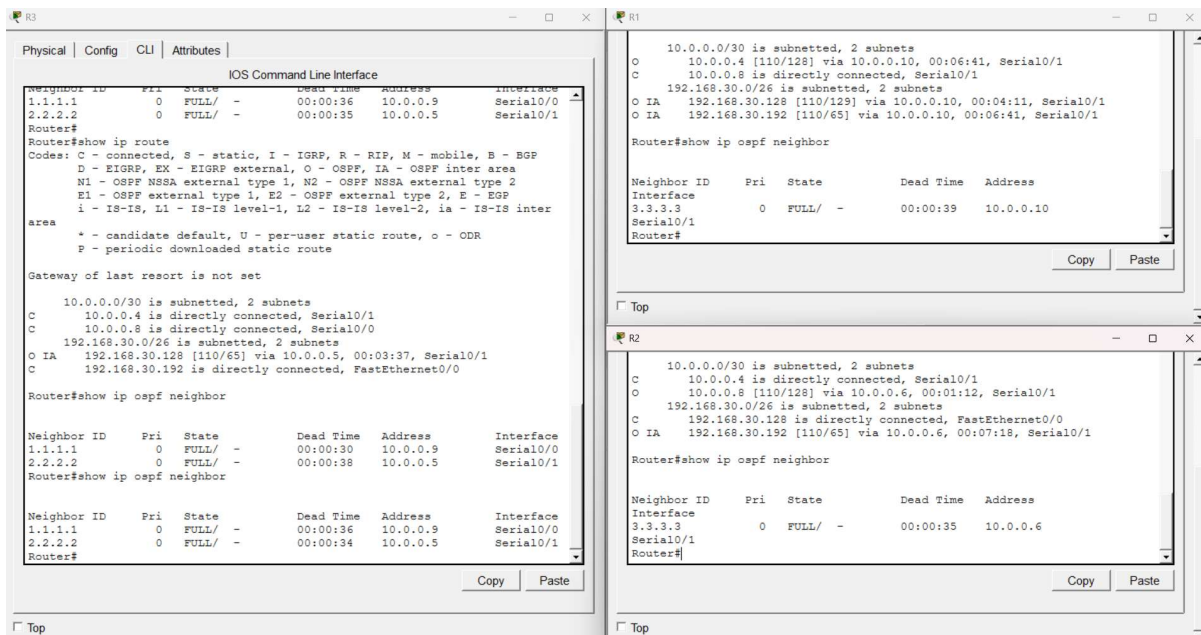


Figure 5: After shutting down the R1-R2 link, R1 and R2 each show only one OSPF neighbor (R3). R3's routing table confirms it now provides transit between R1 and R2, with inter-area routes ('O IA') still reachable via the alternate path.

This demonstrates OSPF's self-healing behaviour: because the topology was a triangle rather than a straight line, losing the direct R1-R2 link did not isolate either router or its LAN - traffic simply rerouted through R3 with a slightly higher path cost.

The R1-R2 interface was then re-enabled with 'no shutdown'. The console log on R1 confirmed the recovery sequence step by step:

- %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
- %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/0 from LOADING to FULL, Loading Done

'show ip ospf neighbor' on both R1 and R2 then showed the direct adjacency re-established: R1 once again listed both 2.2.2.2 (R2) and 3.3.3.3 (R3) as FULL neighbors, and R2 listed both 1.1.1.1 (R1) and 3.3.3.3 (R3). R3, which was never on the failed link, showed both neighbors throughout and was unaffected by either the failure or the recovery.

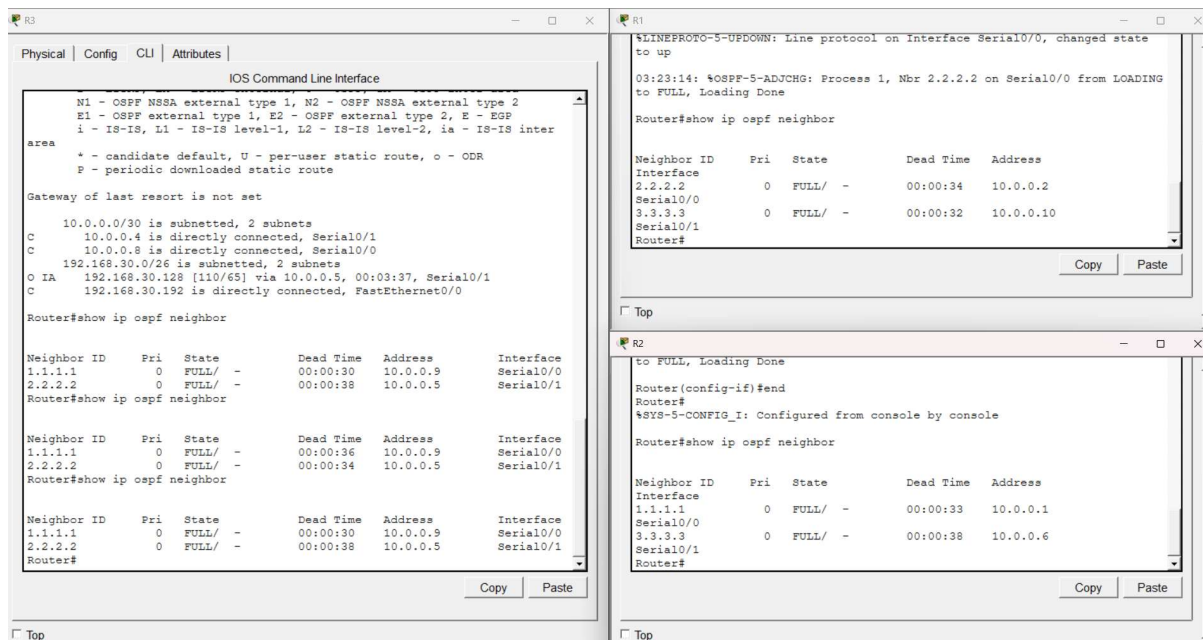


Figure 6: Recovery after 'no shutdown' on the R1-R2 link. R1's log shows the interface coming back up and OSPF re-forming the adjacency with R2 (LOADING to FULL). Both R1 and R2 now show two FULL neighbors again, confirming the direct backbone path was restored.

This confirms that OSPF reconverges automatically in both directions - on failure and on recovery - without any manual route configuration. The dead-timer/hello mechanism detected the link loss quickly, and the LOADING-to-FULL adjacency sequence on recovery shows OSPF re-exchanging its link-state database with the returning neighbor before resuming full routing via the direct path.